PREDICT-THEN-OPTIMIZE FRAMEWORK FOR PUBLIC TRANSPORT LINE REDESIGN UNDER FLUCTUATING TRAFFIC CONDITIONS

Zihao Guo, Andrea Araldo*, Fayçal Touzout, Mounim El-Yacoubi

Institut Polytechnique de Paris, Palaiseau 91120, France (Guo, Araldo, El-Yacoubi); National School of State Public Works, Marne-la-Vallée 77454, France (Touzout)

zihao-eric.guo@ip-paris.fr; andrea.araldo@telecom-sudparis.eu;
faycal.touzout@univ-eiffel.fr; mounim.el_yacoubi@telecom-sudparis.eu

* Corresponding author

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Abstract: Public Transport (PT) lines are traditionally designed to optimize performance under nominal traffic conditions. This paper proposes a predict-then-optimize method to proactively redesign PT networks under traffic fluctuations, balancing passenger travel time, operating cost, and structural continuity. Forecast inaccuracies are sampled from errors of a deep learning model. Computational results on Mandl's benchmark and the Beijing network show improvements while limiting topological modifications. On the Beijing network under high variability, average travel time improves by up to 25.8% while preserving over 85% line overlap; unlike stop-skipping baselines, the redesign preserves full origin–destination connectivity.

Keywords: Public transport network redesign; Predict-then-optimize; Multi-objective optimization; Travel time uncertainty; Structural continuity

1 Introduction

Public Transport (PT) networks are typically planned for nominal, scheduled conditions (Ibarra-Rojas et al., 2015), while real-world operations are continuously subject to variability caused by congestion and other factors (Büchel & Corman, 2020; Danaher et al., 2020), especially in mixed traffic environments. Such routine perturbations occur daily and can propagate across the network, progressively degrading service reliability (Marra & Corman, 2020).

Preventive strategies enhance robustness ex ante through structural measures, such as dedicated bus lanes that isolate buses from mixed traffic (Russo et al., 2022). However, limited street space and high infrastructure costs often make such measures infeasible in dense urban areas (J.-Q. Li et al., 2009). Reactive strategies attempt, at the operational level, to restore service after perturbations have occurred, for instance via stop-skipping, holding, or detouring (Peled et al., 2021; Tang et al., 2018; Zheng et al., 2023). A key limitation shared by both preventive and reactive strategies is that the PT network structure typically remains fixed, despite persistent changes in operating conditions. Network design decisions are revised infrequently, and traditional models assume deterministic or fixed travel times (Büchel & Corman, 2020; Fonseca, 2020), limiting their ability to address recurrent traffic fluctuations.

The main contribution of this paper is to assess the potential of predict-then-optimize proactive redesign of bus networks. To this aim, we propose a new method that adapts bus lines to time-varying stochastic traffic congestion. We solve a multi-objective problem to minimize operational cost and user travel time, and to limit structural network modifications, so as to ease operations and user routing. A possible

deployment rule triggers adaptation only when forecasted conditions sufficiently deviate from nominal conditions; a redesign can be announced in advance (e.g., 1h) to avoid disrupting current journeys. The experiments evaluate a single forecast-conditioned redesign; rolling operation and the trigger are not empirically evaluated.

2 Methodology

2.1 Predict-then-optimize framework

Our methodology follows a predict-then-optimize framework that explicitly integrates current observations into prediction. The prediction model is first trained offline on historical data to learn travel times on road links. During operation, it receives current observations from the latest time steps and outputs a short-term prediction of link travel times $\hat{t}_{ij}$, which are then supplied to the optimization model for potential network redesign. The resulting network design is then evaluated against realized travel times t_{ij} to assess the actual performance of our adaptive network under realistic conditions. To ensure timely responsiveness, this procedure is repeated periodically, i.e., once per hour. At the start of each time slot, the new network is announced in advance (e.g., 1h), so passengers can plan a route that remains valid for their entire trip and is never changed mid-journey; we assume here that the longest trip is shorter than one time slot (1h).

One possible monitoring rule compares the forecast with nominal link travel times. Let E be the substrate edge set and t^0_{ij} the nominal time of link (i, j). The mean relative forecast deviation is

$$\bar{\Delta} = \frac{1}{|E|} \sum_{(i,j)\in E} \frac{|\hat{t}_{ij} - t^0_{ij}|}{t^0_{ij}}. \tag{1}$$

If $\bar{\Delta}$ exceeds a chosen tolerance, an operator may solve the redesign problem for the next slot; otherwise the announced network is retained. The tolerance and operator approval are deployment choices. This rule is proposed as a possible operational mechanism and is not tested as a multi-hour policy in the experiments.

2.2 Network design and objectives

We represent the road infrastructure as a graph with candidate stops and links. Each line is a simple path on this graph and is assigned vehicles subject to a fleet budget. Passengers are assigned to shortest paths on a virtual graph that represents line-specific travel and waiting times. The model requires all considered origin–destination (OD) pairs to remain connected.

Vehicles operating on line ℓ are assumed to traverse its path in both directions, with symmetric travel times. If t^ℓ is its one-way travel time and n^ℓ vehicles are assigned, its headway is $h^\ell = 2t^\ell/n^\ell$. Assuming random passenger arrivals, average waiting time is $w^\ell = h^\ell/2$. A transfer to another line adds that line's expected waiting time and a fixed five-minute penalty. The virtual graph makes these boarding and transfer costs explicit when evaluating passenger paths.

We formulate our public transport network adaptation problem as a three-objective optimization. The first objective, Z_1, minimizes passenger travel time deterioration, where q_k represents the travel demand flow. The use of $\max(\hat{t}_k - t^0_k, 0)$ guarantees that only travel-time increases relative to the nominal benchmark are penalized, thereby avoiding situations where improvements for certain OD pairs compensate for deterioration in others. The second objective Z_2 minimizes operational costs through total vehicle-hours. The third objective Z_3 minimizes network structural changes using the Jaccard index, calculated as the ratio between the number of shared edges (i.e., those present both in the original and in the re-designed network) and the total number of edges (i.e., the union of the edges in the two networks). Structural continuity is handled as an objective (Z_3). Structural continuity means keeping the redesigned lines similar to the lines already in service, so that passengers and operators need to adapt to fewer route changes. For

example, if the original and redesigned networks each have 10 line-labelled links and share 8, their union has 12 links: the Jaccard overlap is $8/12$, and $Z_3 = 1 - 8/12$. A smaller Z_3 means greater overlap and proportionally less change. It is an objective, not a fixed minimum-overlap constraint, so its trade-off with travel time and cost is shown by the Pareto solutions.

$$\min(Z_1, Z_2, Z_3) = \left(\sum_k q_k \max(\hat{t}_k - t_k^0, 0),\ \sum_\ell n^\ell 2\hat{t}^\ell,\ 1 - \frac{\sum_{\ell,\{i,j\}} \min(x_{ij}^\ell, x_{ij}^{\ell,0})}{\sum_{\ell,\{i,j\}} \max(x_{ij}^\ell, x_{ij}^{\ell,0})} \right). \tag{2}$$

Here $\hat{t}_k$ is the predicted shortest travel time for OD pair k on the redesigned network; t_k^0 is its travel time on the original network under nominal link times. For the Jaccard term, x_{ij}^ℓ denotes use of physical link $\{i, j\}$ by line ℓ, and $x_{ij}^{\ell,0}$ its original value. Each line-labelled physical edge is counted once, irrespective of path orientation. Here n^ℓ is the vehicle allocation and $\hat{t}^\ell$ the predicted one-way line travel time; Z_2 is a design-level vehicle-time proxy. When solving our optimization problem, the realized link travel times t_{ij} are not available; we optimize based on predictions $\hat{t}_{ij}$ and then evaluate ex post using realized travel times.

The path and fleet constraints can be expressed compactly as follows. Let N be the stop set, e_{ij}^ℓ indicate the directed path arc from i to j, y^ℓ indicate line activation, a_i^ℓ and b_i^ℓ identify its terminals, and u_i^ℓ be a stop-order variable. Binary variables encode arcs, terminals and activation; n^ℓ is an integer and $\bar{n}$ its per-line upper bound. For each line and candidate stop or arc,

$$\sum_i a_i^\ell = \sum_i b_i^\ell = y^\ell, \qquad a_i^\ell + b_i^\ell \le 1, \tag{3}$$

$$\sum_j e_{ji}^\ell + a_i^\ell = \sum_j e_{ij}^\ell + b_i^\ell, \qquad \sum_j e_{ij}^\ell,\ \sum_j e_{ji}^\ell \le 1, \tag{4}$$

$$u_i^\ell - u_j^\ell + 1 \le |N|(1 - e_{ij}^\ell), \qquad 1 \le u_i^\ell \le |N|, \tag{5}$$

$$e_{ij}^\ell + e_{ji}^\ell \le y^\ell, \qquad y^\ell \le n^\ell \le \bar{n} y^\ell, \quad \sum_\ell n^\ell \le V_{\max}. \tag{6}$$

The first two lines select distinct terminals and enforce flow balance and degree limits. The ordering inequality applies to both candidate directions and eliminates directed subtours. The last line links edge and integer vehicle assignments to line activation and enforces the fleet budget. For each physical link, $x_{ij}^\ell = e_{ij}^\ell + e_{ji}^\ell$; the original-network value is defined analogously.

Passenger routes are not independent network-design variables. For a fixed design, the virtual graph contains line-specific nodes and arcs for travel, access and transfers; waiting time depends on the vehicle allocation. Shortest-path calculations yield the travel time of each OD pair. The model assumes uncongested passenger assignment: it does not model crowding or passenger-flow-dependent travel times.

2.3 Solution algorithm

Given the tri-objective nature of the problem and the nonconvex trade-offs, we adopt an NSGA-III framework (Deb & Jain, 2014) to approximate the Pareto front. The evolutionary search operates on network design variables (line topologies e_{ij}^ℓ and fleet allocations n^ℓ). Each individual encodes a complete PT network design, comprising a set of line path representations and the associated vehicle allocations. Like genes in a chromosome, each line and its assigned vehicles form a block: exchanging or modifying these blocks explores different network designs. NSGA-III can search these discrete designs while retaining diverse trade-offs among the three objectives. At each generation, offspring are produced through selection, variation, and evaluation, followed by the NSGA-III reference-point-based environmental selection, which prioritizes non-dominated solutions while maintaining diversity along the Pareto front. Feasibility-oriented repair addresses line-level structural infeasibility. For each candidate, shortest paths on a virtual-node graph incorporate in-vehicle and waiting times and provide the OD travel times used in the objectives.

Initial networks are constructed from scratch or by locally perturbing the existing routes. Candidates are screened for simple-path validity, connectivity of the considered OD pairs, and a minimum initial Jaccard similarity $J_{\min} = 0.5$. This initialization threshold should not be confused with an enforced overlap threshold for every subsequent solution: structural continuity is an optimization objective. The crossover exchanges complete lines between parents as the primary recombination mechanism; any subsequent modifications are limited to feasibility restoration. Whenever a line topology is inherited from a parent, its vehicle allocation is inherited jointly, thereby preserving the coupling between bus route geometry and service frequency. With probability P_m, a mutation perturbs a single line ℓ via local structural modifications, including node insertion, node deletion, or segment reconnection. Feasibility-oriented repair restores invalid line structures and adjusts fleet allocations; it is not guided by objective improvement. The experiments use crossover probability 0.85, mutation probability 0.35, and 125 reference points. The algorithm returns non-dominated alternatives rather than a single universally optimal design.

3 Evaluation method

Our optimization uses predicted link travel times $\hat{t}_{ij}$, while actual link travel times t_{ij} are unknown at decision time. We therefore evaluate whether the generated line plans remain effective in the presence of prediction errors. We use the Diffusion Convolutional Recurrent Neural Network (DCRNN) (Y. Li et al., 2017) as a representative spatio-temporal predictor to generate empirical one-hour-ahead prediction errors. The objective is not to benchmark forecasting models, but to expose the line-plan generation method to realistic forecast errors. Log-relative errors $\eta_{ij} = \log(\hat{t}_{ij}/t_{ij})$ are sampled with replacement from empirical errors; simulated forecasts are $\hat{t}_{ij} = t_{ij}\exp(\eta_{ij})$. Realized link times follow a log-normal model. For Mandl, the median is fixed at the nominal link time as cv changes. We use the median rather than the mean to retain a typical nominal time while allowing right-skewed travel times and occasional severe delays; the mean consequently increases with cv.

Due to the absence of historical traffic data for Mandl's Swiss network, the DCRNN is trained on the METR-LA dataset (Y. Li et al., 2017), which contains traffic speed measurements from 207 sensors in Los Angeles County collected at 5-minute intervals over four months. For the real-world Beijing case study, instead, we utilize the Q-Traffic dataset (Liao et al., 2018), which provides traffic speed and road network data for the 6th Ring Road in Beijing, China. The transfer of forecast-error patterns from Los Angeles to Mandl is an experimental approximation; pooled residual sampling does not represent link-specific heterogeneity or spatio-temporal correlation.

Mandl's Swiss network, initially used by Mandl (1980), consisting of 15 nodes, 21 links, and a symmetric demand of 15,570 passenger trips, is employed as a canonical benchmark for comparison with existing PT network design methods (Ahern et al., 2022; Liang et al., 2025). To further assess the method's applicability under realistic operating conditions, we consider a real-world urban public transport network in a region in Beijing, China, recently analyzed by Wu et al. (2024), which comprises 301 bus stops, 3,758 road sections, and 29 bus lines. Its origin–destination demand matrix is constructed from smart card data collected during peak hours by aggregating trips between boarding station i and alighting station j. Key parameters are set as follows: a fixed dwell time of $d = 1$ min per stop is assumed following Chen et al. (2022). For Mandl's network, the maximum fleet size is capped at 40, following the service-frequency setting of Ahern et al. (2022) to maintain comparable headways. For the Beijing network, the maximum is set to 150 to accommodate a constant 12-minute headway across this large PT system. We consider coefficients of variation $cv = 0, 0.05, 0.25, 0.50, 1.00$, where cv is the standard deviation divided by the mean; the stochastic experiments use 30 realizations of link travel times and prediction errors. The optimization uses predicted times, and designs are evaluated using realized times. A compromise solution is selected from the Pareto front by its Euclidean distance to the component-wise median objective vector.

The Mandl experiments compare redesigned networks with static route-frequency solutions and a frequency-only adjustment, in which line structures remain fixed while vehicle allocation changes. The Beijing comparison uses the original network and stop-skipping variants. A one-hour maximum

computational time per instance is used for the offline experiments; this is a computational budget, not evidence that a complete operational deployment can be carried out within one hour. We use three indicators. IS_{20} is the percentage of OD pairs whose travel time is at least 20% lower than in the reference network; each OD pair counts equally. OCR is the percentage reduction in the operating-cost proxy $\sum_\ell n^\ell 2t^\ell$ relative to the reference network; positive values indicate savings. LO is the number of shared line-labelled edges divided by the number in either network, expressed as a percentage: 100% means no edge change. The reported values are means $\pm$ standard deviations over 30 realizations.

4 Numerical results

4.1 Mandl benchmark

We use predicted travel times as input to the optimization problem, and we evaluate the resulting adaptive network against realized link travel times and prediction errors across multiple scenarios. As travel-time variability increases, both 20% Improvement Share (IS_{20}) and Operational Cost Reduction (OCR) rise systematically, reflecting the increasing value of adaptive redesign under highly fluctuating conditions. Across all variability levels and redesign schemes, Line Overlap (LO) remains high, indicating that these improvements result from targeted adjustments rather than wholesale restructuring. Table 1 compares redesigns of four to twelve lines with the static route-frequency solutions of Ahern et al. (2022). For the four-line instance, at $cv = 0$, redesign achieves $IS_{20} = 5.79 \pm 2.99\%$, OCR $= 10.20 \pm 5.31\%$ and LO $= 84.34 \pm 5.44\%$. At $cv = 1$, these become $30.57 \pm 11.60\%$, $39.20 \pm 12.94\%$ and $77.26 \pm 5.19\%$, respectively. Thus, against Ahern et al., more OD pairs gain at least 20% as variability increases, while the reported redesigns retain substantial line overlap.

The frequency-only baseline improves service under variability, but it captures substantially less of the OD-level improvement than full redesign; for example, at $cv = 1$, its 20%-improvement share is $16.47 \pm 9.53\%$, compared with $30.57 \pm 11.60\%$ for our method. While frequency-only adjustment keeps the line topology fixed, the proposed line redesign can reroute service away from deteriorated links. In a forecast-error sensitivity analysis, realized times are held fixed while the error scale varies: zero gives a perfect forecast, one reproduces empirical error, and larger values stress the model. The results indicate that the opportunity for redesign grows with variability and that the Pareto front changes little across the tested error scales.

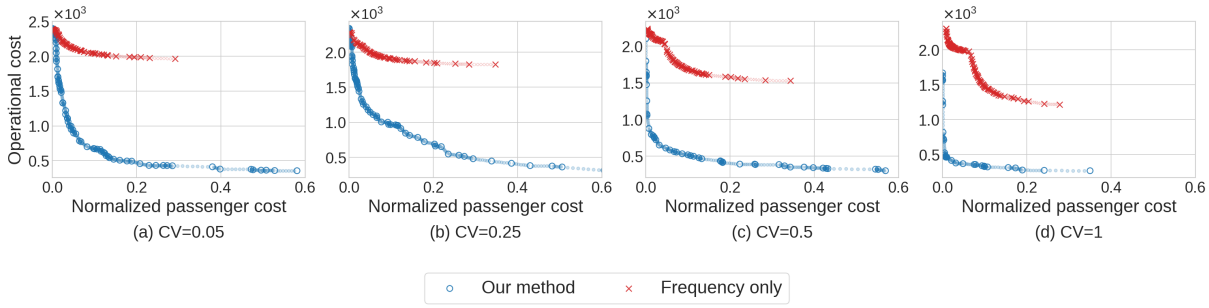


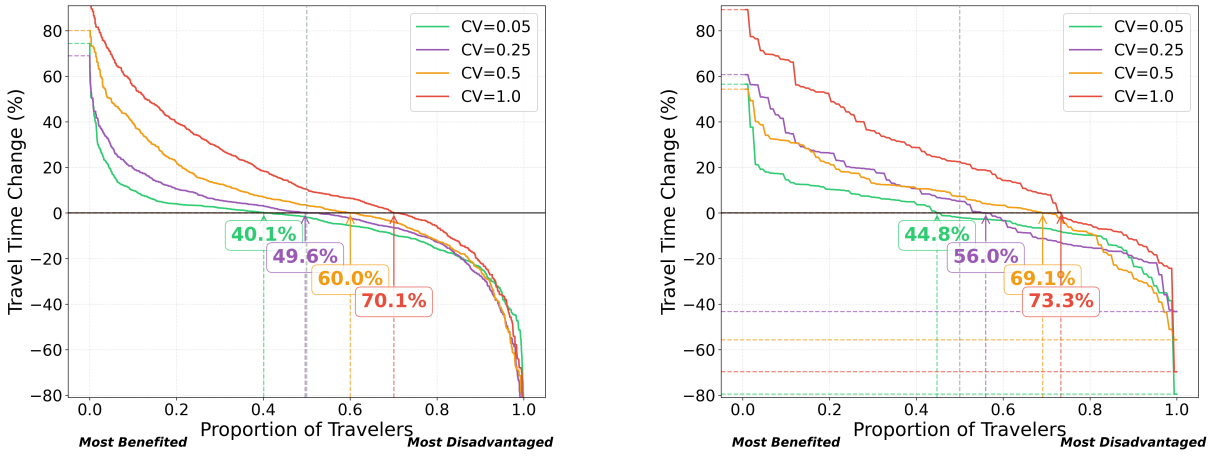
Figure 1: Travel-time and cost (Z_1, Z_2) Pareto-front projections for the four-line Mandl instance under four travel-time variability levels.

Figure 1 plots the two-dimensional projection of the Pareto front, highlighting the trade-off between normalized passenger travel-time deterioration Z_1 and the operating-cost proxy Z_2. Each point maps to a specific network design, with the lower-left corner representing the preferred direction for both minimization objectives. The third objective, structural continuity, is not shown in this projection; line overlap for the redesigned networks is reported in Table 1.

Figure 2 shows trip-time changes in one randomly chosen realization.

Table 1: Performance of redesign compared with the static solutions of Ahern et al. (2022) on Mandl's network. Values are mean ± standard deviation over 30 realizations; CV is the coefficient of variation.

Lines	Metric	CV=0	CV=0.05	CV=0.25	CV=0.50	CV=1.00
4	IS_{20} (%)	5.79 ± 2.99	8.25 ± 4.65	15.40 ± 5.66	24.63 ± 7.91	30.57 ± 11.60
	OCR (%)	10.20 ± 5.31	13.28 ± 7.37	16.86 ± 6.74	24.06 ± 12.01	39.20 ± 12.94
	LO (%)	84.34 ± 5.44	83.79 ± 7.73	78.83 ± 6.65	77.53 ± 5.27	77.26 ± 5.19
6	IS_{20} (%)	8.96 ± 4.47	9.53 ± 3.60	16.31 ± 5.10	23.46 ± 7.84	31.50 ± 10.46
	OCR (%)	25.94 ± 4.71	28.94 ± 5.40	32.10 ± 5.69	36.75 ± 9.55	48.90 ± 10.19
	LO (%)	79.58 ± 2.29	81.20 ± 2.80	78.16 ± 3.94	79.63 ± 4.47	79.16 ± 4.08
8	IS_{20} (%)	11.52 ± 2.67	13.99 ± 3.90	18.62 ± 5.64	26.84 ± 8.76	35.71 ± 11.53
	OCR (%)	26.97 ± 4.45	29.92 ± 3.84	31.51 ± 5.27	35.35 ± 7.98	45.35 ± 9.23
	LO (%)	83.58 ± 4.07	84.95 ± 3.07	82.51 ± 4.03	80.80 ± 5.25	81.14 ± 4.23
10	IS_{20} (%)	8.83 ± 3.15	9.74 ± 4.54	14.27 ± 5.61	21.46 ± 8.88	31.21 ± 11.56
	OCR (%)	33.62 ± 4.18	34.65 ± 4.93	37.23 ± 5.53	40.62 ± 8.71	52.59 ± 9.17
	LO (%)	77.57 ± 3.40	78.14 ± 2.76	76.96 ± 3.22	78.77 ± 4.09	78.34 ± 3.46
12	IS_{20} (%)	9.76 ± 3.57	10.34 ± 3.19	14.74 ± 3.51	21.44 ± 6.96	32.44 ± 11.91
	OCR (%)	35.94 ± 4.95	37.51 ± 5.54	39.56 ± 5.18	44.20 ± 6.90	54.27 ± 9.00
	LO (%)	80.69 ± 2.98	80.95 ± 3.40	80.61 ± 2.78	81.46 ± 3.27	80.77 ± 4.11



(a) Ahern et al.

(b) Vermeir et al.

Figure 2: Trip travel-time distributions under different variability levels in one randomly chosen realization. Panels compare redesign against the static networks of Ahern et al. (2022) and Vermeir et al. (2021). Positive values indicate shorter trips under redesign.

4.2 Beijing network

We compare our method against a stop-skipping operational baseline on the Beijing network. For each variability level, links whose predicted travel time exceeds the original by more than a preset threshold are classified as degraded. The stop-skipping strategy then removes intermediate stops adjacent to degraded links whenever a direct shortcut is available, while leaving the overall line structure and vehicle allocation unchanged. This baseline represents a simple, local response to travel-time disruption, but structurally constrained: it cannot reroute vehicles or reassign fleet, and skipping stops may disconnect part of the demand at a stop from the network entirely.

The results in Table 3 show that the original network exhibits some inefficiencies: the solution of our optimization yields an 11.40% average travel-time improvement even at $cv = 0$. This reflects static re-optimisation of the incumbent network rather than prediction-driven adaptation. The benefit grows

Table 2: Beijing results at $cv = 1$ compared with the original network. Travel-time improvement for stop-skipping is calculated on reachable OD pairs.

Method	Overlap (%)	Cost reduction (%)	Travel-time improvement (%)	Unreachable OD (%)
Stop-skipping, 25% threshold	42.18	12.71	14.71	38.93
Stop-skipping, 50% threshold	47.83	12.39	19.71	35.28
Stop-skipping, 100% threshold	55.88	13.49	15.11	28.05
Redesign	85.42	22.56	25.78	0

with variability. Table 2 exposes a basic tension in stop-skipping: operational benefit comes at the cost of connectivity. Aggressive thresholds improve travel time on the reachable network but disconnect a large share of OD pairs as variability grows. Because no stops are removed, all OD pairs stay reachable under the redesign. Travel-time improvements for stop-skipping and redesign are calculated on different reachable OD sets and should therefore be interpreted alongside the connectivity figures.

Table 3: Redesign results on the Beijing network across traffic variability levels, compared with the original network.

Coefficient of variation	0	0.05	0.25	0.50	1.00
Line overlap (%)	90.08	87.34	86.51	86.39	85.42
Operational cost reduction (%)	0.50	1.53	3.85	12.02	22.56
Average travel-time improvement (%)	11.40	11.60	16.99	18.44	25.78

Table 3 shows the reported trade-off across all five settings. Average travel-time improvement is computed as the reduction in the sum of OD travel times relative to the original network, divided by the corresponding original sum; it is not the fraction of OD pairs improved by 20%, nor a demand-weighted average. Operational cost reduction reflects the model's vehicle-time proxy, which excludes costs such as crew scheduling and depot operations.

5 Conclusion

This paper systematically studied the potential of proactive bus line redesign, targeted at limited parts of the network, in maintaining user and operator performance in the case of severe traffic fluctuations. We formulated a multi-objective optimization problem balancing passenger travel time, operational costs, and network stability. Our predict-then-optimize approach relies on an NSGA-III-based optimization framework that proactively redesigns bus lines based on predicted line travel times. Experiments on Mandl's network and a real-world Beijing network demonstrate effective adaptation across varying levels of travel-time uncertainty. Overall, the gains grow with travel-time variability and come with high line overlap, indicating that the benefit does not rely on large structural change.

A natural extension is to enrich the forecast-error model with explicit spatial and temporal correlation, allowing network-wide common shocks to be represented. A further direction is to incorporate in-vehicle crowding, for instance through a capacitated or congestion-dependent passenger assignment; this would let the redesign account for the load it induces on each line and report crowding-aware rather than purely uncongested travel-time gains.

Deployment also requires advance notice, passenger information, and operator coordination; these practical costs and passenger responses are not evaluated here.

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